Effects of Feedback Valence and Task-Difficulty Priming on Task Enjoyment

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Abstract

This study extends earlier investigations of task difficulty priming and feedback valence influencing individuals' task enjoyment. Tang and Sarsfield-Baldwin (1991) suggest that expectations about task difficulty and the nature of performance feedback can influence how individuals experience and evaluate an activity, even when the task itself remains unchanged. Building on this framework, the present study examined how task difficulty priming and feedback valence interact to influence task enjoyment in an online setting. We hypothesized that positive feedback would increase enjoyment ratings, that easy task priming would lead to greater enjoyment than hard priming, and that positive feedback would be especially effective when paired with a hard task prime. Using a 2 (Prime: Easy vs. Hard) × 2 (Feedback: Positive vs. Neutral) between-participants design, 82 participants completed an anagram-solving task administered through Qualtrics and subsequently rated their enjoyment on a 5-point Likert scale. Descriptive statistics and a two-way analysis of variance were used to assess main effects and interactions between priming and feedback. Although patterns in the means suggested differences in enjoyment across conditions, the analyses revealed no statistically significant main effects of prime or feedback, nor a significant interaction between the two variables. The present study's findings contrast with earlier laboratory-based work demonstrating robust effects of task labeling and feedback on task liking and intrinsic motivation. The results highlight the importance of considering how motivational processes may differ across operational definitions and research contexts.

Keywords: priming, feedback valence, task enjoyment, experiment, motivation, liking, task difficulty

Shaping Task Enjoyment

Understanding what influences individuals' enjoyment of a task or activity has long been a central topic in psychological research. A common, and logical, assumption is that performance feedback directly shapes how much individuals enjoy a task: positive comments should make an activity feel more rewarding, whereas neutral feedback should dampen or maintain the same level of enthusiasm. Yet, scientific studies suggest that task enjoyment is influenced by numerous interacting factors, including individuals' expectations about the task difficulty, the context in which feedback is received, and motivational processes that operate before and during task engagement (Tang & Sarsfield-Baldwin, 1991). Before even completing a task, people may form biased expectations about their own ability based on subtle cues, such as whether the task is described as "easy" or "difficult" (Tang & Sarsfield-Baldwin, 1991). These expectations can substantially shape subsequent task engagement and emotional responses.

Decades of research demonstrate that perceptions, expectations, and contextual clues can influence people's experiences of effortful activities. For example, labeling a task as difficult can lead individuals to expect themselves to be less likely to succeed in completing it (Tang & Sarsfield-Baldwin, 1991). As a result, participants assigned to difficult tasks may put in less effort in completing tasks and could feel less enjoyment. Conversely, an easy-labeled task may foster confidence in individuals and lead to higher expectations of success, potentially enhancing enjoyment (Tang & Sarsfield-Baldwin, 1991). Consequently, the present study's question arises: how does feedback valence interact with task difficulty expectations to shape task enjoyment?

Tang and Sarsfield-Baldwin (1991) examined how task context and evaluative cues interact with individual differences to shape task enjoyment and motivation. Using a $2 \times 2 \times 2$ design, they manipulated self-esteem (high vs. low), task label (difficult vs. easy), and

performance feedback (positive vs. negative) while participants completed identical anagram tasks consisting of 25 anagrams. Although the tasks themselves did not differ in objective difficulty, labeling them as “easy” or “difficult” altered participants’ expectations of success, which in turn influenced their affective responses to the task. Overall, positive feedback increased task liking and intrinsic motivation, consistent with theories of intrinsic motivation and self-efficacy (Deci & Cascio, 1972; Bandura, 1987). Importantly, the effects of feedback depended on task labeling and self-esteem: under difficult task labels, individuals with high self-esteem increased task liking following positive feedback, whereas individuals with low self-esteem showed this increase primarily under easy task labels. These findings suggest that task enjoyment is not driven by feedback alone, but by the interaction between perceived task difficulty, evaluative feedback, and individuals’ expectations about their own competence.

Although Tang and Sarsfield-Baldwin’s (1991) work provides important insights, their studies were conducted in controlled laboratory settings, relied on bogus performance feedback delivered face-to-face, and measured intrinsic motivation through free-choice observation. These methodological features differ from contemporary research environments, where tasks are frequently administered online, feedback is delivered automatically, and subjective enjoyment is often measured through self-report scales. As a result, it remains unclear whether the motivational processes identified by Tang and Sarsfield-Baldwin generalize to modern, unsupervised digital contexts. Such limitations in generalizability are one of the main points the present study aims to address.

The Present Study

The present study extends Tang and Sarsfield-Baldwin’s (1991) findings by focusing specifically on the construct of task liking and examining how task difficulty priming and

feedback valence influence enjoyment in an online setting. Rather than incorporating self-esteem and behavioral measures of intrinsic motivation, this study isolates task label priming (easy vs. hard) and feedback type (positive vs. neutral) to examine their effects on self-reported task enjoyment. The present study tests the generalizability of and expands Tang and Sarsfield-Baldwin's (1991) work to modern digital research environments by testing whether similar effects emerge when the task is delivered entirely online, without an experimenter physically present, and when enjoyment is measured using self-report.

This experiment not only replicates key components of Tang's paradigm but also tests them with modified definitions, using neutral feedback instead of negative feedback and measuring task enjoyment with a 5-point Likert scale instead of a 7-point scale. Neutral statements were chosen because we were theoretically interested in whether minimal to no affirmation is sufficient to sustain or reduce enjoyment. The present study examines whether these motivational processes generalize to online survey contexts, where participants complete tasks in unsupervised, naturalistic environments and are not given practice anagrams prior to participation. By shifting to a Qualtrics-based procedure and focusing on subjective enjoyment, our study contributes to understanding how contextual cues operate in everyday digital settings.

Our primary question is how task difficulty priming and feedback valence impact task enjoyment. We hypothesized that positive feedback would be associated with higher task enjoyment than neutral feedback. Additionally, we hypothesized that tasks primed as easy would elicit higher enjoyment than tasks primed as hard. Finally, we hypothesized an interaction such that participants primed to believe the task was hard would report higher enjoyment when receiving positive feedback compared to participants primed to believe the task was easy.

Method

Participants

A total of 119 participants were recruited using convenience sampling with a snowball recruitment approach. Recruitment primarily involved undergraduate students at Northwestern University, as well as the researchers' friends and family. Individuals who received the Qualtrics survey link were asked to distribute it to their peers. Participants did not receive compensation for their participation and were informed that they could discontinue the survey at any time without penalty. All participants completed the task electronically in an unsupervised setting of their choosing. It should be noted that 37 entries were removed from the data collection due to repeated responses from the same individuals, which were indicated by repeated IP addresses. The final sample consisted of usable data collected from 82 participants.

Materials

The study was conducted using a survey made on Qualtrics. All participants were shown five anagrams in the same order as listed in Table 1. The anagrams were adapted from Ekstrand and Dominowski (1968) and Tang and Sarsfield-Baldwin (1991). All five anagrams were selected to be two syllables because, according to Muncer and Knight (2010), the number of syllables in an anagram is a major factor in determining its difficulty, with more syllables indicating greater difficulty. Therefore, a uniform number of two syllables was set to select the anagrams. Though all five anagrams were of the same level of difficulty, the anagrams were displayed as either hard or easy. The hard and easy labels were part of the manipulation of the experiment. For instance, for CANOE, participants were shown "(HARD: 1 Solution)" under the directions "Rearrange the letters in the following word to make a new word."

Table 1

Five anagrams shown to participants with their corresponding solution(s).

Anagram	Solution(s)
Diver	Drive, Rived
Table	Bleat
Canoe	Ocean
Rebuy	Buyer
Mixer	Remix, Mirex

Participants were given a text box underneath each anagram question to type in their answer. There was no visible timer counting down the seconds, but participants were informed that they were given 90 seconds to answer each question. When the 90 seconds to solve the anagram were up, the survey automatically led participants to the next page, where they were presented with either a positive or neutral statement, regardless of how accurate their responses were. Participants in the positive feedback group received messages such as “Wow, that's really good!” while those in the neutral feedback group were only shown “Okay!” Feedback statements were adapted from Corpus and Lepper (2007).

Lastly, participants were shown a page to rate their enjoyment of the anagrams activity. On the five-point Likert scale, participants were presented with five buttons laid out horizontally, with the leftmost button labeled “Dislike a great deal”, second leftmost button labeled “Dislike somewhat”, the middle button labeled as “Neither like nor dislike”, the second rightmost button labeled “Like somewhat”, and the rightmost button labeled as “Like a great deal”. Though not labeled as 1-5, the task enjoyment rating data were collected as 1 corresponding to “Dislike a great deal” and 5 corresponding to “Like a great deal.”

Procedure

The experiment used a 2 (Prime: Easy vs. Hard) × 2 (Feedback: Positive vs. Neutral) between-participants factorial design. Task enjoyment was measured using a Likert-type enjoyment rating. Participants were randomly assigned to four conditions: easy priming with positive feedback, easy priming with neutral feedback, hard priming with positive feedback, or hard priming with neutral feedback. Participants completed the Qualtrics survey on their electronic devices in an unsupervised setting of their choice. Participants were sent the link to the Qualtrics survey by a member of the research team. Before solving the anagrams, participants were presented with a definition and explanation of what an anagram is, in case they were not familiar with anagrams. Upon beginning the questionnaire, participants were asked if they were willing to participate in the study, with the option to exit the questionnaire at any time. Participants were then shown directions reading, “In the following study, you will be asked to solve a series of short anagrams. You will have 90 seconds to complete each anagram before being moved to the next one.”

In the survey, participants were shown anagrams presented with the descriptions of the anagrams’ difficulty level and number of solutions for each anagram. Participants were presented with the message "the following is a series of [easy/hard] anagrams.” After solving each anagram, participants were shown a page containing either a positive or neutral feedback message. After the feedback statement, participants clicked the “→” to move to the next page and solve the next anagram, and so on. After solving all five anagrams, participants were required to answer a question indicating how much they enjoyed the activity (solving the anagrams) using the aforementioned task enjoyment rating scale. It should be noted that due to a Qualtrics malfunction, participants in the easy-positive condition completed only four anagrams instead of the standard five. All other conditions completed the full set of five anagrams.

Results

Task Enjoyment Ratings

The statistical analysis was conducted using the software RStudio.

Table 2

Descriptive Statistics for Task Enjoyment as a Function of Priming and Feedback

Feedback	Prime					
	Easy			Hard		
	M	*SD*	*SE*	*M*	*SD*	*SE*
Positive	3.50	1.24	0.36	4.31	0.88	0.17
Neutral	4.05	1.15	0.26	4.08	0.88	0.18

Note. M, SD, and SE represent mean, standard deviation, and standard error, respectively.

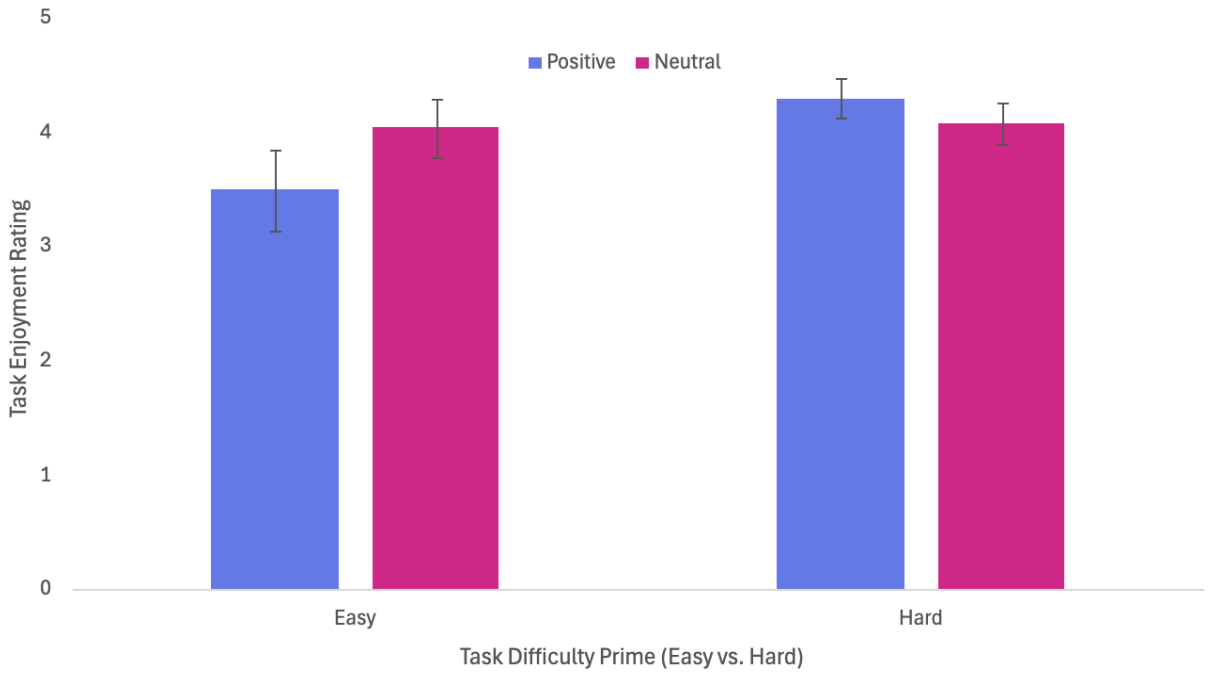


Fig. 1. Mean task enjoyment ratings as a function of task difficulty prime (easy vs. hard) and feedback type (positive vs. neutral). Enjoyment ratings were slightly higher for hard-primed anagrams than for easy-primed anagrams in the positive feedback condition. In contrast, enjoyment ratings in the neutral feedback condition were very similar across easy and hard primes. Error bars represent standard errors of the mean.

A 2 x 2 ANOVA was conducted for Prime (Easy vs. Hard) and Feedback (Positive vs. Neutral). The analysis revealed that there was no significant main effect of prime, $F(1,78) = 3.26$, $p = 0.08$. The results also revealed there was no significant main effect of feedback, $F(1,78) = 0.49$, $p = 0.49$. The interaction between prime and feedback was also not significant, $F(1,78) = 2.76$, $p = 0.10$.

It should be noted that a 2 x 2 ANOVA of the non-excluded data with 119 participants revealed that there was a significant main effect of prime, $F(1,115) = 6.61$, $p<0.01$. However, the analysis revealed that there was no significant main effect of feedback, $F(1,115) = 0.57$, $p = 0.45$. The ANOVA revealed that there was no significant interaction between prime and feedback, $F(1,115) = 1.38$, $p = 0.24$. A t-test was conducted to clarify where the significant main effect of prime came from. A two-sample t-test (of the non-excluded data) comparing the task enjoyment ratings between the easy-positive and hard-positive groups within the positive feedback condition showed that participants who received positive feedback enjoyed the task significantly more when the anagrams were labeled “hard” than when they were labeled “easy,” $t(53) = -2.48$, $p<0.05$.

Discussion

The present study examined whether task difficulty priming (easy vs. hard) and feedback valence (positive vs. neutral) influence task enjoyment in an online anagram task. Drawing on

prior work by Tang and Sarsfield-Baldwin (1991), we hypothesized that positive feedback and easy task priming would independently increase task enjoyment and that positive feedback would be especially beneficial when tasks were framed as difficult. Contrary to these predictions, the results revealed no significant main effects of priming or feedback, and no significant interaction between the two variables. Although these findings did not support our original hypotheses, they could still contribute meaningfully to theoretical accounts of task enjoyment and motivation.

These findings suggest that, in the present study, priming task difficulty and brief performance feedback were insufficient to reliably alter task enjoyment. This partially diverges from Tang and Sarsfield-Baldwin's (1991) findings, which demonstrated that performance feedback and task labels meaningfully shaped task liking and intrinsic motivation, particularly when moderated by individual differences such as self-esteem. One possible explanation is that Tang's effects depended on the interaction between two different societies (students in China and the United States) and personality variables, whereas the present study isolated task label and feedback without incorporating measures of individual differences. Without such moderators, the influence of priming and feedback on enjoyment may be weaker or less consistent. Additionally, Tang and Sarsfield-Baldwin (1991) distinguished between task liking measured via self-report and intrinsic motivation measured behaviorally, noting that these outcomes can reflect different cognitive and motivational processes. The present study relied solely on a self-report measure of enjoyment, which may be less sensitive to subtle motivational shifts than behavioral measures. Such contrasts may help explain why the present study's results did not replicate the stronger effects observed in Tang and Sarsfield-Baldwin's (1991) laboratory-based work.

Three important confounds should be considered when interpreting the present findings. First, repeated survey responses were identified and removed from the dataset, resulting in a final sample of 82 participants. Notably, analyses conducted before these exclusions indicated a significant main effect of task prime; however, this effect likely reflected noise introduced by repeated responses rather than a true experimental effect. Excluding these responses reduced statistical power and may have limited the ability to detect small but meaningful effects. Second, a technical malfunction in Qualtrics resulted in participants in the easy–positive condition completing only four anagrams instead of the intended five. Differences in enjoyment for this group could plausibly be attributed to reduced task duration or effort rather than to priming or feedback manipulations. Shorter tasks may be perceived as less effortful or more enjoyable, complicating the interpretation of condition differences involving this group. The last confounding variable concerns the unsupervised nature of the online study. Because participants completed the task in a setting of their choosing, they may have used external resources (e.g., search engines or AI) to assist with solving the anagrams. Access to outside help could have altered both task performance and subjective enjoyment, making the task less reflective of participants' actual problem-solving abilities. For some participants, external assistance may have increased enjoyment by reducing effort and allowing quicker task completion, whereas for others it may have reduced enjoyment by diminishing the sense of personal accomplishment. This variability introduces noise into the measurement of task enjoyment and may have weakened the observed effects of priming and feedback.

Beyond these confounds, several limitations should be considered when interpreting the findings of the present study. First, task enjoyment was operationalized using a single self-report Likert scale, which may not fully capture motivational engagement or effort. It is possible that

self-reported liking and observable behavior do not always align, and therefore, future studies could reflect different operational definitions of task enjoyment, such as the time taken to solve the tasks. Perhaps a participant could take longer to complete the tasks because they are uninterested or unmotivated, or could take less time to complete the tasks because they are enjoying the activity and are enthusiastic about completing more. Second, the study employed neutral rather than negative feedback, which may have reduced the contrast between feedback conditions and limited the strength of the manipulation. For a stronger manipulation, future studies could use negative feedback statements like they were used in Tang and Sarsfield-Baldwin (1991). However, such an approach would need to be carefully considered due to potential ethical concerns related to participant well-being.

Despite these limitations, the present study contributes to understanding how motivational cues operate in contemporary digital environments. Online learning platforms, educational applications, and corporate training modules frequently rely on task descriptions and automated feedback to shape user experience. The current findings suggest that such cues may not uniformly influence enjoyment when delivered in brief, unsupervised contexts. Future research could incorporate behavioral indicators of engagement, manipulate objective task difficulty, or examine how individual differences moderate feedback effects in online settings.

All in all, although the present study did not replicate the significant effects reported by Tang and Sarsfield-Baldwin (1991), it nonetheless provides valuable insight into the boundary conditions of feedback and priming effects on task enjoyment. By adapting classic motivational manipulations to a modern online format, this research underscores the need to consider how theoretical constructs translate across different contexts and measures. Continued investigation

along these lines will help clarify when and how contextual cues meaningfully shape motivation and enjoyment in real-world digital environments.

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